

Laser scanners in ROS2

Let's now do something a little less theoretically intensive. We are going to have a look at a laser scanner being simulated by a professional robotic simulator, **MVSim**, within the infrastructure of ROS2.

As you know, ROS2 allows for the intercommunication of different *nodes* (executables) through a system of *topics* and *messages*. When using a simulator node like **MVSim**, this node will be responsible for handling the calculations that generate the response of the sensors, and the result of this process will be published as a message on a specific topic. Then, any node which wants to use the sensor response can access it simply by subscribing to the topic and receiving the corresponding messages.

Launching an example

If you have a working installation of ROS2 Humble on Ubuntu, you can install MVSim with

```
sudo apt install ros-humble-mvsim
```

Let's launch a simple demo that comes pre-configured with MVSim:

```
ros2 launch mvsim demo_1robot.launch.py
```

You will see two windows open: one is the GUI of the simulator itself (MVSim), and is part of the same executable. The other window, *Rviz*, is an all-purpose visualizer for ROS2. This window is **not** part of the MVSim executable, and the only way it can display the map, the robot's pose and the laser scanner readings is because it is subscribing to the relevant topics and getting the information through the ROS2 pipeline.

Inspecting the data

Let's try and figure out how this communication is happening between MVSim and Rviz. Write below the commands necessary to:

- Check which topics are being published.

```
ros2 topic list
```

- See which nodes are publishing/subscribed to the laser scanner topic/topics.

```
ros2 topic info /scan
```

- Print to terminal the value of the laser scanner messages that are being published.

```
ros2 topic echo /scan
```

This is all well and good, but, of course, we would like to see the result of the simulation when something a bit more interesting is happening. Using the previous commands to see which topics exist, can you figure out how to make the robot move?

Hint: **Once you know the topic that you want to use, you can use the `rqt` tool to publish the messages, as we saw in the last session.

```
ros2 topic pub /cmd_vel geometry_msgs/msg/Twist "{linear: {x: 0.3, y: 0.0, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 0.3}}"
```

In []: